

Welcome to the notebook for the ML course. Its sole goal is to help you understand the material of each lecture better both for the intends and purposes of the final exam of the course and also many of the contents adhere to the current ml-related industry standard junior level interview questions.

The material of the notebook in order to be fully conquered would require you to dedicate one to two hours maximum per day for 2 or 3 days. But if you are a weekend warrior two four hour study intervals will get you to a satisfactory degree of preparation. I would advise against the latter approach though but you have to plan for contingencies.

Regarding access to the material, I need your gmail account to grant you permissions to be able to view it. The pdfs mentioned later will be available through my github repository page (link below). The github repository also includes my weekly assignments from 2024 and at the beginning of each they have an ai generated overview (that was added after the course) that may help you understand why you had to use what you were asked to.

Below is a study guide to help you roadmap your approach to get the most out of the material it is the optimal way to study it as it was designed this way:

- Take a deep look into the presentations for each lecture. They all contain the material of the lectures plus better qualitative explanations and use case examples. The latter ones also contain the answers to some of the questions towards the end of the finals question list as they are more complex and its better to take a look at them during the presentation first.
- After you parse through the lectures take a look at the dedicated presentation of the kernel trick. It is the most mathematically abstract topic of the course but once you understand its inner workings, subdivisions (linear and non linear) you will be able to explain it with ease if asked to do so.
- Now when you are done with the presentations its time to dive in the exam topics answers. There are two different approaches present in the pdfs. The analytical ones and the qualitative ones.
- Start with the analytical ones (Questions x-y Finals). The analytical ones explain in great depth the answers to the exam questions. Keep in mind that they are way more detailed in comparison to your teacher's expectations. Do not be frightened if you see extended mathematical notation as you will not be asked to show mathematical derivations during the exam. The analytical explanations are for you to still gain more intuition regarding the inner workings and use cases of the topics presented in the course. Read them once maybe twice if you something is still unclear to you, there is no use to memorising them.
- Now for the important part, if you followed the roadmap so far the final step is to parse through the qualitative explanations of the exam questions (Qualitative Questions x-y Finals). These pdfs contain qualitative answers for all the questions present in the finals pdf

in a qualitative manner that showcase to the professor that you understood the materials fundamentals, use cases, dos and donts, and inner workings to an above satisfying degree. Given my experience with Bonfini the theoretical exam's aim is just for you to show that you understood all the material in a qualitative manner and in a way that you could simply explain it to a friend of yours nothing more advanced than that. There are certain exceptions were it is possible for you to be asked to derive mathematical steps towards an objective but dont be anxious about it as I have that covered as well

- Finally there are a few questions that "may" require mathematical derivation. This is the final step towards your studying for the course. There is a detailed pdf (Questions with explicit formulas or derivations) that showcases everything you need in order to provide the mathematics necessary to answer the questions that require you to do so. The most likely candidate from these would be the least squares derivation or the logistic regression.

A bonus final step would be for you to review the weekly assignments you submitted during the semester (or consult the ones in the github page) because after following the roadmap you will be able to understand better why you were asked to use the specified approaches to the problems presented in spite if your submissions were ai-generated. Your goal is to gain a qualitative understanding regarding the use of the topics you were taught.

Also regarding the neural networks in order to understand how backpropagation works in an expert level and in a more "chill way" I would highly recommend that you watch the 4 first videos of this playlist: https://www.youtube.com/watch?v=aircAruvnKk&list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Once you completed your approach based on the roadmap above you will be ready for the exam. If you understood and learnt the Qualitative Questions you will most likely get a 9-10 grade.

If you have any questions submit them to the email fukatasdiavasma@gmail.com I will try to reply as soon as I can. Also spread this message to the rest of the class because I dont have the gmails of everyone (only gmail will work). I wish I could share this sooner but better now than never.

gitub link: <https://github.com/basiliswork/Introduction-to-Data-Science-And-Machine-Learning-I>

Cheers

Vasilis